

Closure Taxonomy: What Closure Is (and Is Not)

Amos Jay Maley

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Abstract

The term *closure* is routinely misapplied in scientific and philosophical discourse, often conflating restrictions on inference with claims about explanatory exhaustion or ontological completeness. This paper introduces a strict taxonomy of closure types and establishes that only one—*move closure*—is admissible under standing- and admissibility-preserving discipline. Explanatory and ontic closures are shown to be structurally inconsistent. The result blocks totalization errors while preserving open inquiry wherever admissible witnesses, scopes, and failures remain.

1 Scope Fixation

This paper permits the following classes of questions:

- What classes of inferential moves are eliminated by admissibility enforcement?
- What forms of inquiry remain legitimate after such elimination?
- How must “closure” be typed to avoid illicit totalization?

The following questions are forbidden:

- Does closure imply the end of discovery?
- What explanations replace those ruled out?
- Can closure itself be explained, optimized, or grounded in deeper causes?

This paper classifies *inferential permissions*. It does not adjudicate truth-claims, world-states, or metaphysical commitments.

2 Role Assignment

Anchors

- Admissibility
- Standing

- Fail-closed enforcement
- Non-derivability of anchors

Anchors are fixed by structural necessity. Any attempt to motivate, derive, vary, or optimize anchors constitutes illicit anchor motion and is inadmissible.

Tensors

- Closure operators (elimination of inadmissible moves)
- Scope and envelope boundaries
- Standing-preserving redescription constraints

Tensors transmit constraint across domains without introducing explanation.

Skin

- Ordinary language of “closure,” “completion,” and “finality”
- Rhetorical expectations of explanatory exhaustion
- Sociological interpretations of theoretical maturity

Skin elements are explicitly non-authoritative and carry no binding force.

3 Structural Analysis

The term *closure* is structurally overloaded. This paper distinguishes three non-equivalent types. Only one is admissible.

3.1 Type I — Explanatory Closure (Inadmissible)

Explanatory closure asserts that no further explanations are possible.

Such claims attempt to terminate inquiry by asserting explanatory sufficiency. They require either:

- derivation or justification of anchors, or
- explanation of why explanation itself must end.

Both moves violate admissibility constraints and collapse standing.

Status: Structurally inconsistent. Rejected.

3.2 Type II — Ontic Closure (Inadmissible)

Ontic closure asserts that the space of what exists is fully enumerated.

This form converts limits on inference into claims about being. It performs illicit scope transport from epistemic constraint to ontology, treating absence of explanation as metaphysical exhaustion.

Status: Structurally inconsistent. Rejected.

3.3 Type III — Move Closure (Admissible)

Move closure eliminates specific illicit inferential moves, including:

- anchor derivation or variation,
- metaphor treated as mechanism,
- totalizing unification,
- post-hoc repair of standing failure.

Move closure operates solely on *permission to infer*. It does not restrict events, entities, outcomes, or future discoveries. It constrains discourse without asserting completeness.

Status: The only admissible form of closure.

4 Failure and Closure Results

This paper closes the following recurrent failures:

1. **“Closure means nothing new can happen.”** This confuses elimination of illicit inference with exhaustion of contingent events.
2. **“Closure means inquiry is finished.”** This substitutes inferential constraint for empirical availability.
3. **“Closure means the framework explains everything.”** Explanation is precisely what is not permitted at the anchor level.
4. **“Closure is a claim about reality itself.”** Closure governs discourse, not ontology.
5. **Soft totalization.** The claim that all meaningful inquiry must terminate within the framework is explicitly rejected. Closure restricts *how* questions may be asked, not *which* questions may arise as witnesses and scopes change.

All failures arise from collapsing move closure into explanatory or ontic closure.

5 Non-Generative Termination

No new explanations are introduced. No metaphysical commitments are made. No future results are implied.

The paper terminates with the invariant:

Closure, properly typed, eliminates illicit moves. It does not exhaust inquiry, enumerate reality, or terminate discovery.